



Max. Marks: 100

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

20 Marks

- 1-

10. Which of the following is a primary difference between Hardwired Control and Microprogrammed Control in generating control signals within the control unit?
- A) Hardwired Control uses software routines to generate signals, while Microprogrammed Control uses fixed logic circuits. B) Hardwired Control is faster but less flexible, while Microprogrammed Control is slower but more adaptable to complex instructions.
- C) Microprogrammed Control is used only in RISC architectures, while Hardwired Control is exclusive to CISC architectures. D) Microprogrammed Control generates signals directly through hardware, whereas Hardwired Control relies on control memory.
11. During the execution of an unconditional branch instruction, what control signal sequence is primarily responsible for updating the Program Counter (PC) with the Branch Target Address (BTA)?
- A) Fetching the next instruction into the Instruction Register (IR). B) Checking conditions before updating the PC with the new address
- C) Decoding the offset value and adding it to the current PC value to calculate BTA. D) Writing the BTA directly to memory without updating the PC.
12. _____ are the different types of generating Control Signals.
- A) Hardwired B) Micro instructions
- C) Microprogrammed D) Both A&C
13. What is the key difference between Synchronous DRAM (SDRAM) and Asynchronous DRAM?
- A) SDRAM is synchronized with the system clock, allowing faster data access compared to Asynchronous DRAM, which operates independently of the clock. B) SDRAM operates without a clock signal, while Asynchronous DRAM uses a clock.
- C) Asynchronous DRAM has a built-in clock, whereas SDRAM does not. D) SDRAM and Asynchronous DRAM have identical performance but differ in size.
14. How many 512K x 8 memory chips are needed to construct a 2M x 32 memory module, and how are they arranged?
- A) 8 chips arranged in 2 rows and 4 columns B) 16 chips arranged in 2 rows and 8 columns
- C) 32 chips arranged in 4 rows and 8 columns D) 16 chips arranged in 4 rows and 4 columns
15. Which type of ROM can be erased using ultraviolet light and then reprogrammed, making it suitable for firmware updates during development but not in normal operation?
- A) PROM (Programmable ROM) B) EEPROM (Electrically Erasable Programmable ROM)
- C) Mask ROM D) EPROM (Erasable Programmable ROM)
16. In computer architecture, which cache mapping technique offers the simplest implementation but can lead to frequent conflicts between memory blocks, thus reducing cache performance?
- A) Fully Associative Mapping B) Direct Mapping
- C) Set-Associative Mapping D) Multilevel Mapping
17. In Computer Architecture, which type of interrupt allows the processor to complete the current instruction before handling the interrupt, ensuring that the execution state is not disrupted?
- A) Precise Interrupt B) Maskable Interrupt
- C) Non-Maskable Interrupt D) Vectored Interrupt
18. Which statement best describes block mode DMA operation?
- A) The DMA controller transfers one data byte at a time, releasing the bus after each transfer. B) The CPU controls each data transfer in block mode, ensuring synchronized operations.
- C) The DMA controller transfers a block of data in a single burst, holding the bus until the entire block is transferred. D) Block mode DMA is only used for small data transfers due to bus contention issues.

19. In distributed arbitration, how is the bus master selected among multiple competing devices?
- A) Devices compete by sending their unique identification numbers, and the one with the highest ID wins the arbitration process.
 B) Each device sends a request to the central controller, which selects the highest priority device.
 C) A round-robin approach is used, where each device takes turns becoming the bus master.
 D) The device with the lowest power consumption automatically becomes the bus master.
20. Which key feature differentiates PCI (Peripheral Component Interconnect) from USB (Universal Serial Bus) in terms of device connectivity and data transfer?
- A) PCI is designed for external peripheral devices, while USB is strictly for internal components.
 B) USB provides faster data transfer rates than PCI for connecting internal hardware components.
 C) PCI and USB both primarily serve the same purpose and are interchangeable in most systems.
 D) PCI supports direct memory access and bus mastering for high-speed internal component connections, while USB is designed for external peripheral connections with hot-swapping capabilities.

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Illustrate the functional units of Computer with the neat diagram.	5	L*1	1	1,2
	b) With a practical example show how instructions are executed sequentially in a computer.	5	L2	1	1,2
	c) With the help of neat diagram explain Sequential Circuit multiplication process by multiplying 13×11	6	L3	2	1,2
2.	a) Define addressing modes. With example and syntax explain any four addressing modes.	6	L1	1	1,2
	b) Apply the Carry Save Addition (CSA) technique to perform the multiplication of +21 and +15. Show the step-by-step process.	4	L2	2	1,2
	c) Multiply the following signed numbers $A = -11$ and $B = 27$ using Booth Algorithm.	6	L3	2	1,2
3.	a) With neat diagram show how processor is interfaced with memory hence explain basic operational concepts.	6	L1	1	1,2
	b) Analyze basic performance equation and computer performance measurements.	4	L2	1	1,2
	c) With the help of neat diagram and algorithm perform $8/3$ using restoring division algorithm	6	L3	2	1,2
Unit – II					
4.	a) Demonstrate three bus Organization of a data path with a neat diagram	6	L1	3	1,2
	b) What is asynchronous DRAM? Model the internal organization of a $2M \times 8$ dynamic memory chip.	6	L2	4	1,2
	c) List and explain the different types of ROM.	4	L3	4	1,2
5.	a) Illustrate and explain the internal organization of a 128-bit RAM structured as a 16×8 cell array, detailing its key components and operation with a diagram.	6	L1	4	1,2
	b) How does control unit generates control signals in case of Hardwired control? Explain it with example.	6	L2	3	1,2

c)	Explain the concept of Translation look aside buffer (TLB) with neat diagram.	4	L3	4	1,2
6. a)	With a block diagram illustrate the basic organization of a microprogrammed control unit	6	L1	3	1,2
b)	Compare and contrast Direct Mapping and Associative Mapping used in cache memory, explaining how each method works and providing examples of their application.	6	L2	4	1,2
c)	Outline the control sequence generated for the following instructions using single bus architecture.				
i)	MOVE R2 (R1)	4	L3	3	1,2
ii)	Add R2, (R1)				

Unit – III

7. a)	What are interrupts? How hard nested interrupts and simultaneous interrupts are handled?	6	L1	5	1,2
b)	Describe Direct Memory Access with the neat diagram.	10	L2	5	1,2
8. a)	Define Exceptions. Describe types of exceptions.	4	L1	5	1,2
b)	Illustrate the concept of Centralized arbitration and distributed arbitration techniques with the neat diagram.	6	L2	5	1,2
c)	Explain the following:				
i.	Universal serial bus (USB)	6	L1	5	1,2
ii.	Peripheral Component Interconnect (PCI) Bus				

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
